Angled and negative work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use force direction to determine work's sign and size.

SUCCESS CRITERIA - I can...

1. Compare force direction with displacement.
2. Use the parallel force component.
3. Explain positive, negative and zero work.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A positive triangular pulse peaks at **50 N** and lasts **0.3 s**. It acts on a **3 kg** cart initially at **-2 m/s**. **Find impulse** and final velocity.

Marzano check - circle: ① ② ③

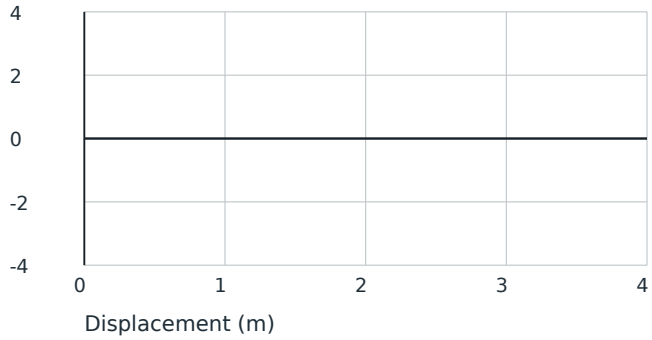
02 Worked example: explain the prediction

Reference: $W = Fd\cos\theta = F_{parallel}\, d$.
θ is between force and displacement. CPA uses the supplied parallel component. Opposite force gives negative work.

Worked reasoning: A box moves **4 m** right while friction acts **3 N** left. Friction's work is **-3×4=-12 J** because force opposes displacement.

Your turn: 1. **Draw force** and displacement arrows. 2. **Add a perpendicular upward force** and **state its work** for this horizontal motion.

Force along path (N)



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The box moves right. Classify work by a rightward pull, leftward friction and upward normal force.

Pull right; friction left



Motion right

The rectangle is the box. Add the requested force and displacement labels.

Marzano check - circle: ① ② ③

04 Apply the physics

A **20 N** rope pulls a crate at **30°** above horizontal as the crate moves **5 m** horizontally. Friction is **4 N**.
1. Draw and **label the force angle** and displacement.
2. **Calculate rope work**, friction work and net work.

Add force and displacement arrows; label the angle.



Force: ____ N Displacement: ____ m

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A force perpendicular to velocity can bend a path without changing speed. Explain using work.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Fragments of masses **4** and **6 kg** separate from rest. The **4 kg** piece moves at **+12 m/s**. **Find the other velocity** and total kinetic energy.
Use before and after snapshots to organize the momentum account.

Before	**After**

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A **10 N** force acts at **120°** to a **2 m** displacement. **Find work** and **explain the sign**.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **20 N** pull at **60°** above horizontal moves a box **3 m** right while friction is **4 N**. **Find work** by pull, friction and their sum.

Add force and displacement arrows; label the angle.

Force: ____ N Displacement: ____ m

Marzano check - circle: 1 2 3